



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Occupancy-Aware Smart Library Seat Reservation with Predictive Demand Analytics

CSA0925-Java programming

Under the Guidness of
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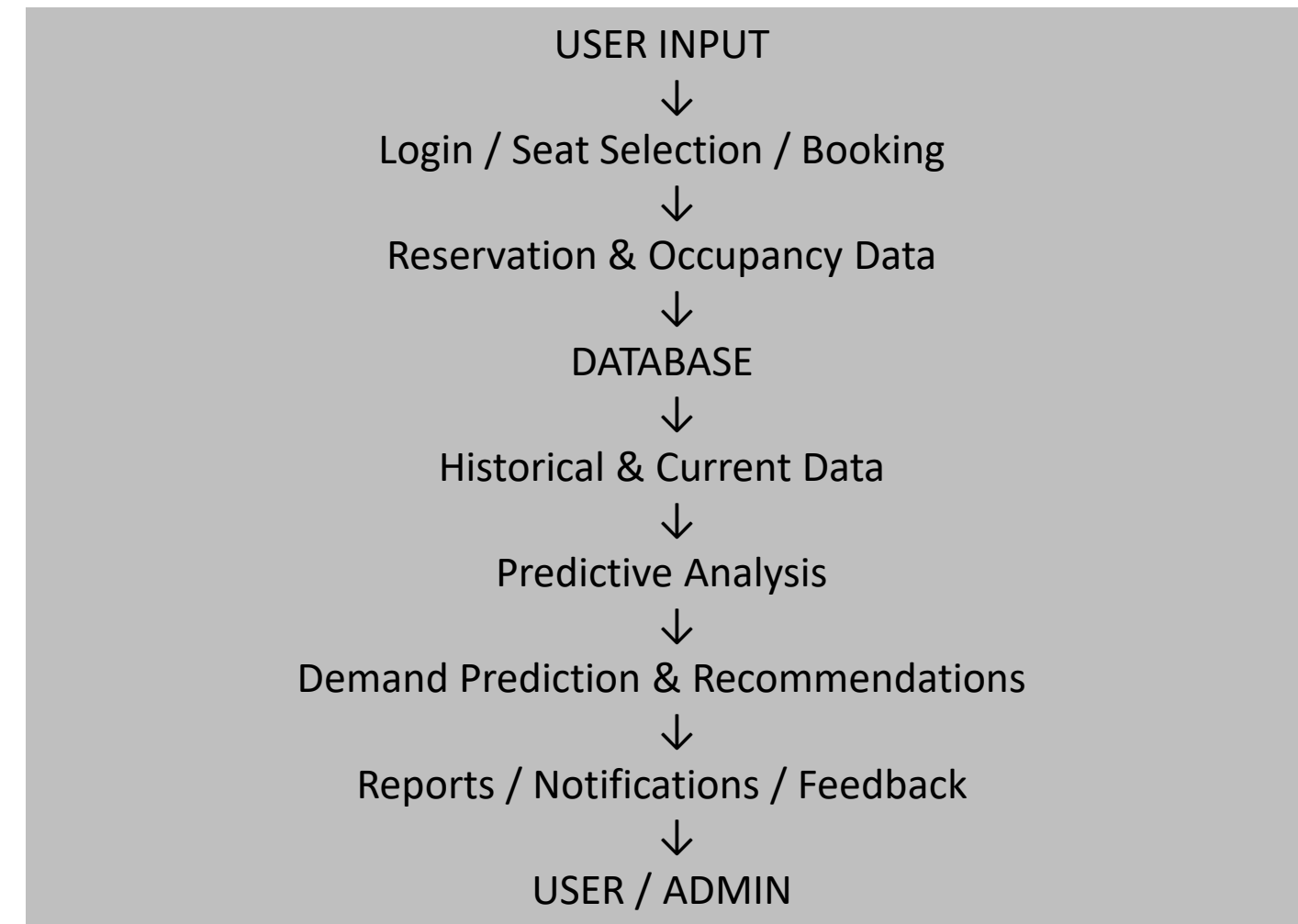
Module Design

Module	Purpose
1. Smart Library Seat Reservation & Occupancy Management	To reserve seats and manage real-time seat availability and occupancy.
2. Predictive Demand Analytics & Smart Recommendation	To predict busy periods and recommend suitable seats and time slots.
3. Tracking, Mentor Feedback & Communication	To track usage, collect feedback, and communicate updates with users.

Interfaces used

User Interface | Reservation & Occupancy Interface | Analytics & Recommendation Interface | Admin Interface

Data Flow

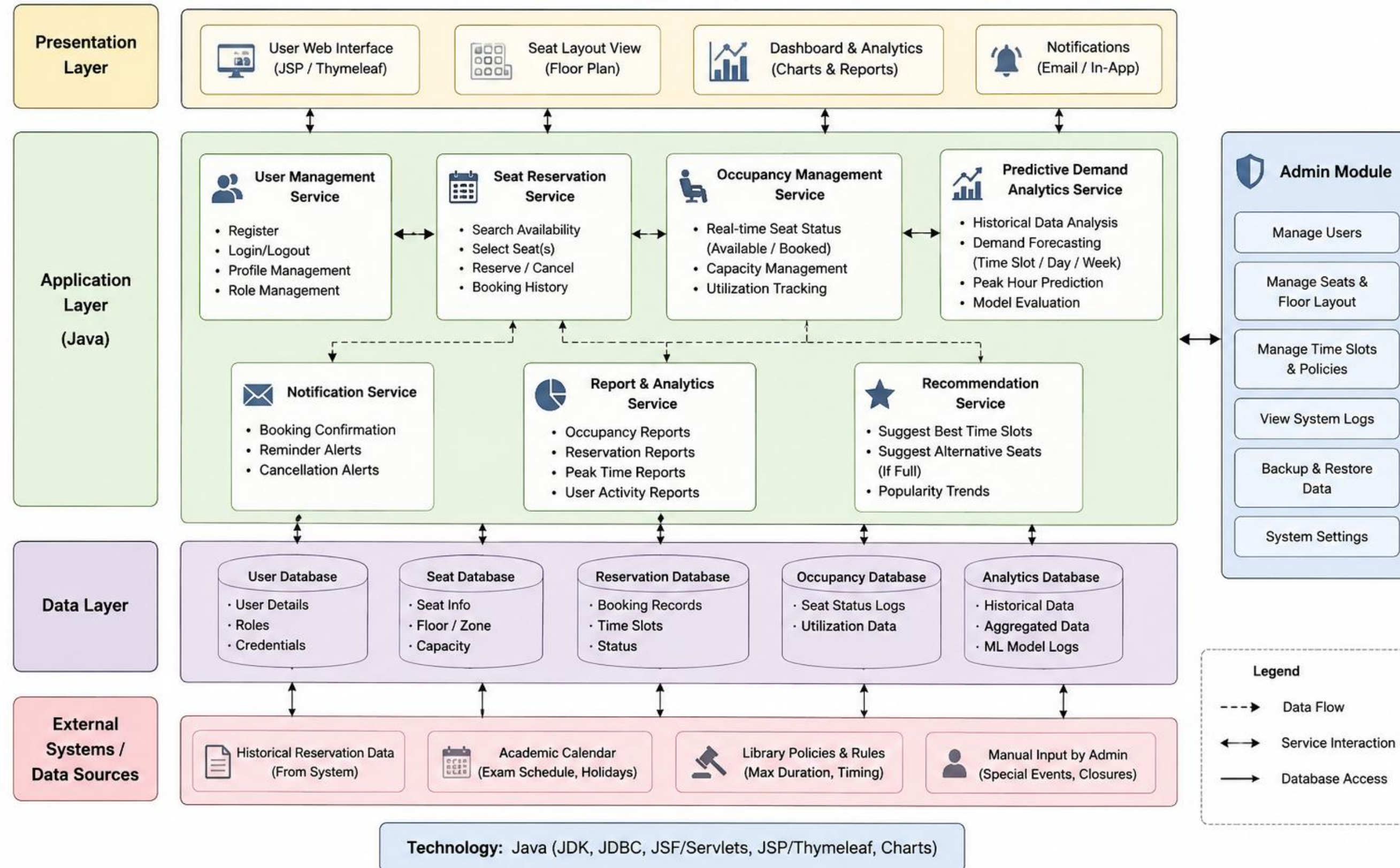




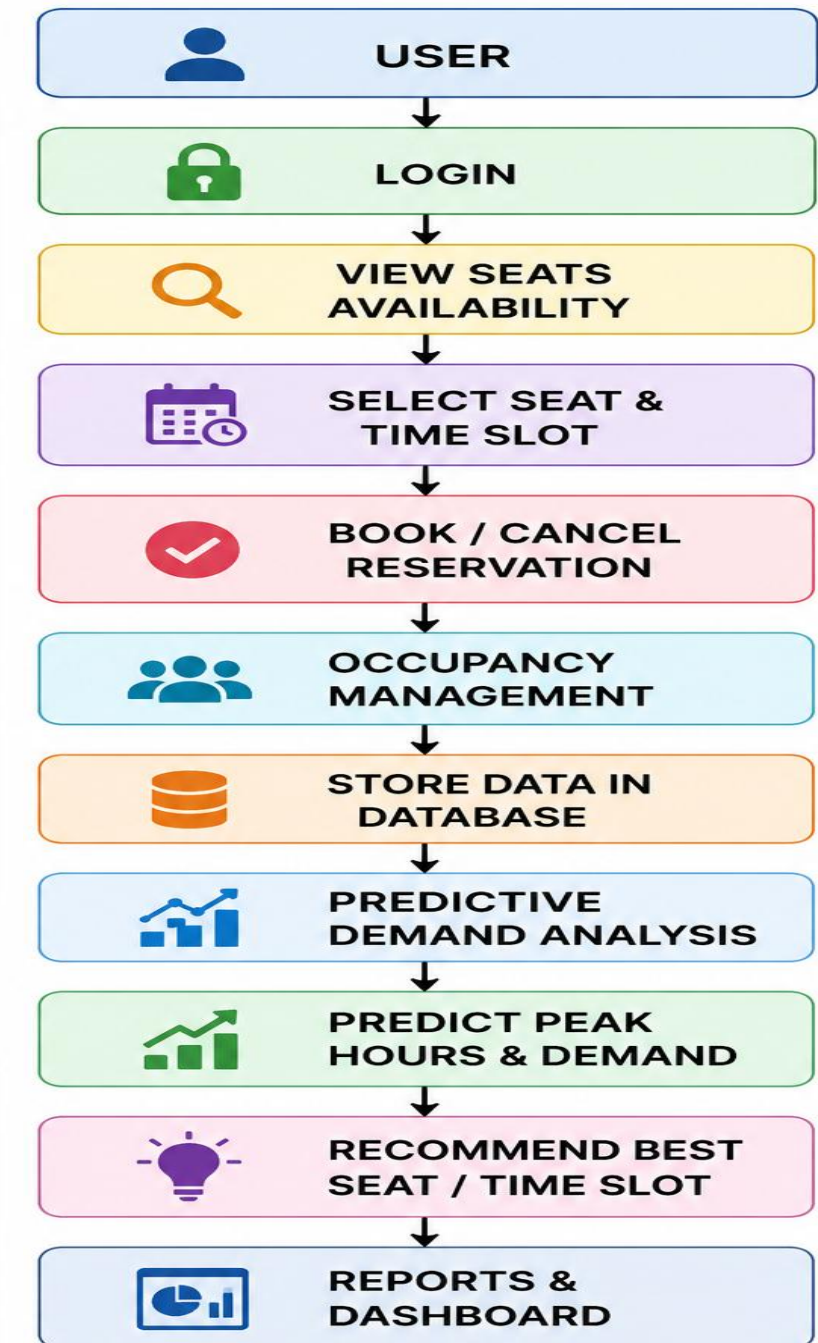
Module	Algorithm	Purpose
1. Smart Library Seat Reservation & Occupancy Management	Greedy Seat Allocation Algorithm	Assigns the best available seat quickly
	Hash Map / Hash Table	Maintains real-time seat availability and occupancy
	FCFS (First Come First Serve)	Handles reservation requests fairly



Updated Architecture



Final Module Flow





UML & DB

Use Case

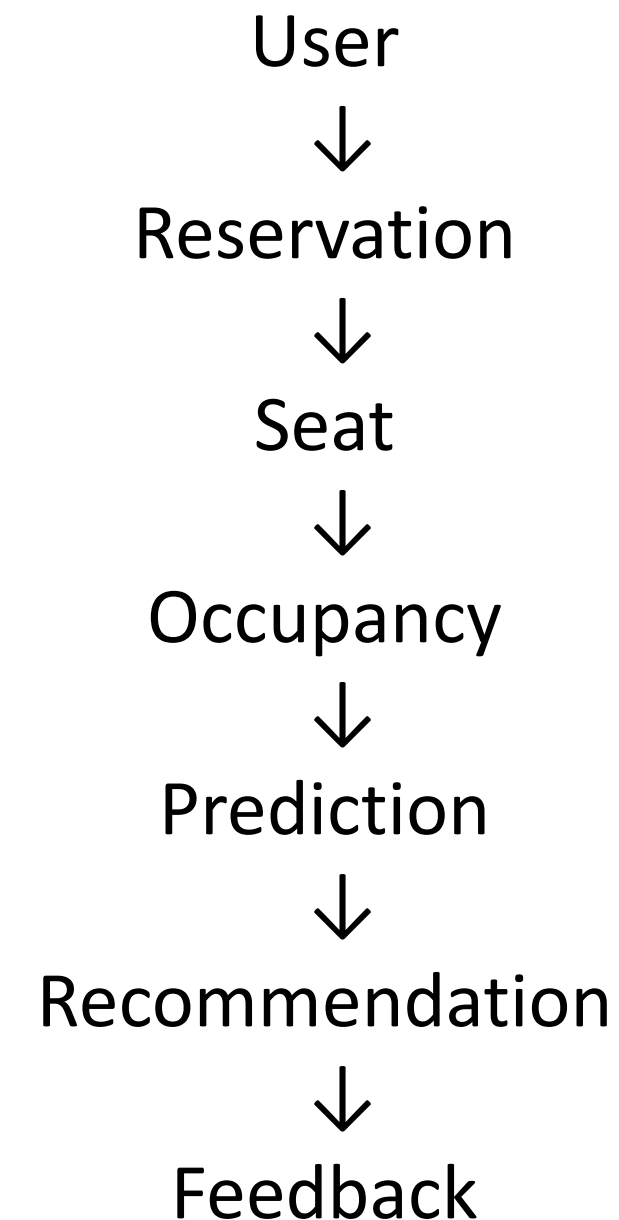
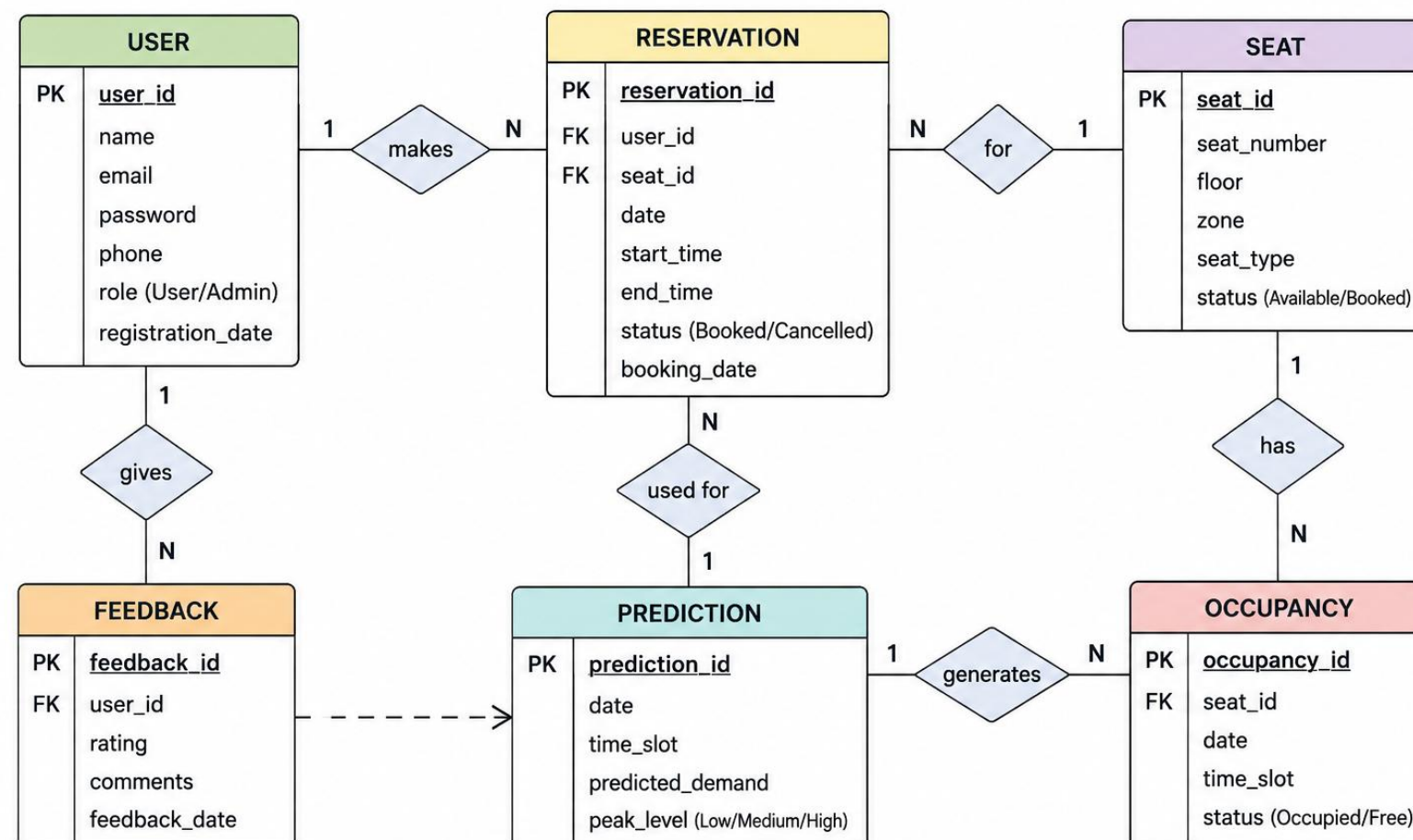
Actors: User, Admin

Main Use Cases: Login, View Seats, Book Seat, Cancel Booking, View Prediction, Get Recommendation, Manage Seats, View Reports.

Sequence

User → Login → View Seats → Select Seat → Book
→ Database → Confirmation

ER DIAGRAM – SMART LIBRARY SEAT RESERVATION SYSTEM





Technology Stock

Languages

- Java
- SQL
- HTML,JS(front design)

Frameworks

- Java Swing / JavaFX
- JDBC

Libraries

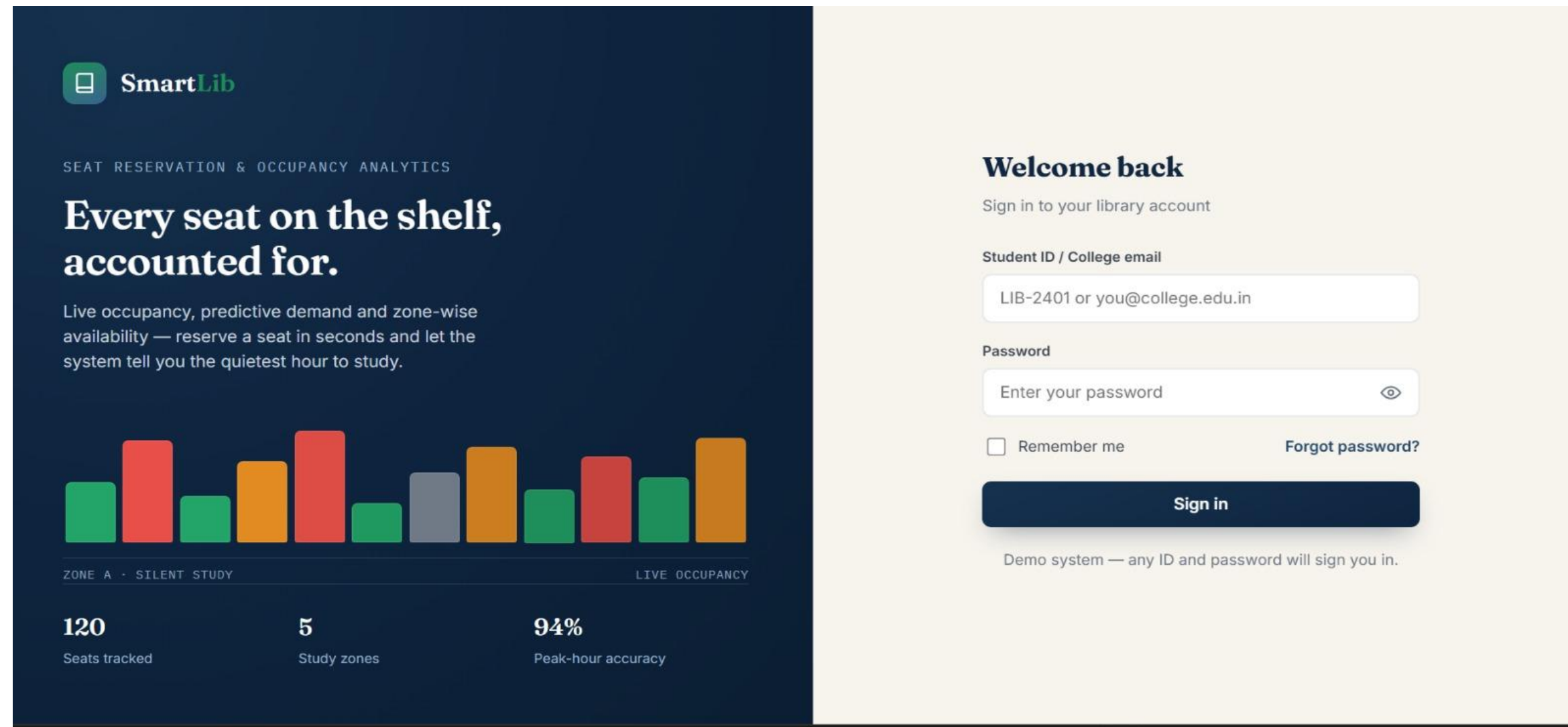
- MySQL Connector/J
- JFreeChart

Hardware

- Computer/Laptop
- Server
- No IoT devices



IMPLEMENTATION



Code summary

A Java-based library seat reservation system that manages bookings, tracks occupancy, and stores data in a database. The system uses historical reservation data to predict demand and recommend suitable seats and time slots.



IMPLEMENTATION

SmartLib

Dashboard

Seat Map

My Reservations

Analytics

Notifications

Profile

ADMIN

Admin Overview

Dark mode

Logout

Search seat, zone, reservation ID...

Mon, 17 Aug, 2026 · IST

Moderate

DS Demo Student LIB-2401

Good to see you, Demo Student

Here's what the library looks like right now.

TOTAL SEATS

120

+5% this week

AVAILABLE

54

Across 5 zones

OCCUPIED

48

Peaks at 2 PM

RESERVED

18

Next 3 hours

Current library occupancy

48 / 120 seats occupied right now

Library status: Moderate

40% occupied

Silent Study82%

General Reading61%

Discussion Area45%

Digital Learning73%

Reference38%

SMART RECOMMENDATION

Your usual 2:00 PM slot tends to run high occupancy on weekdays.

10:00 AM – 12:00 PM

Silent Study, Zone A

~42% occupancy

View available seats

SmartLib

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Logout

Search seat, zone, reservation ID...

Mon, 17 Aug, 2026 · IST

Moderate

DS Demo Student LIB-2401

My reservations

Manage upcoming, active and past bookings.

ID	SEAT	ZONE	DATE	TIME	STATUS	ACTIONS
RSV-1042	A12	Silent Study	18 Aug 2026	10:00–12:00	Upcoming	<button>View</button> <button>Cancel</button>
RSV-1039	C05	Discussion Area	17 Aug 2026	02:00–04:00	Active	<button>View</button> <button>Check out</button>
RSV-1021	B18	General Reading	12 Aug 2026	09:00–11:00	Completed	<button>View</button>
RSV-1010	D07	Digital Learning	08 Aug 2026	01:00–03:00	Cancelled	<button>View</button>



TESTING

Unit Tests

- Test login and user registration.
- Test seat booking and cancellation.
- Test occupancy and prediction calculations.

Integration Tests

- Check Java application ↔ MySQL database.
- Check reservation ↔ occupancy updates.
- Check prediction ↔ recommendation flow.

Outputs

- Successful login/registration
- Seat booking confirmation
- Updated seat availability
- Demand prediction and recommendations
- Reports and notifications



CHALLENGES

Next Plan

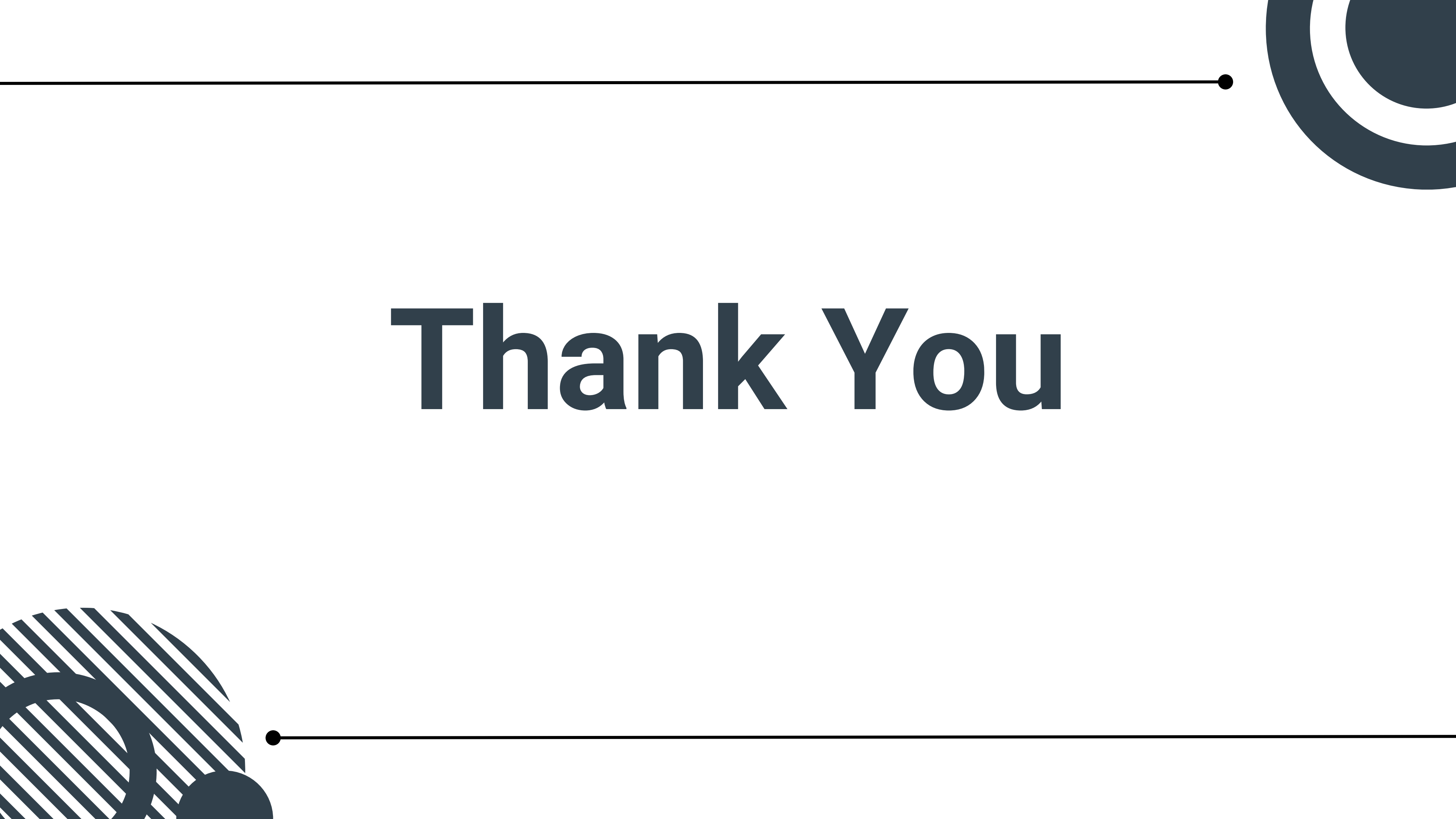
- Improve prediction accuracy.
- Add more analytics and reports.
- Improve the user interface.
- Add more recommendation features.

Issues

- Managing seat availability and booking conflicts.
- Collecting accurate occupancy data without IoT.
- Predicting demand from limited historical data.

Solutions

- Use database validation to avoid double booking.
 - Use reservation and manual check-in/check-out data.
 - Use historical data for demand prediction.
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Thank You